

Qiaoya Wu

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Education

University of Illinois Urbana-Champaign (UIUC) , PhD candidate	Aug 2021 - present
Advisor: Prof. Yue Shen	
Research areas: Quasar spectral surveys; AGN accretion physics and photoionization	
Xiamen University (XMU) , Bachelor of Astronomy	Sep 2017 - Jun 2021
Advisor: Prof. Hao-Ran Yu and Prof. Jianfeng Wu	
Research areas: Cosmological N-body simulations; Black hole binaries	
Thesis: Spin mode reconstruction in Lagrangian space	

Followships and Awards

Chu Excellence in Research Graduate Student Award , UIUC	Apr 2025
Lewis E. Snyder Graduate Student Travel Award , UIUC	Mar 2025
CAPS Graduate Fellowship , Center for AstroPhysical Surveys, UIUC	2022-2023
Outstanding Undergraduate Student Awards , XMU	Jun 2021
Caiwenzhong Fellowship , College of Physics Science and Technology, XMU	Apr 2020
Academic Excellence Scholarship , Department of Astronomy, XMU	Oct 2019
Guangqi Fellowship , Shanghai Astronomical Observatory, CAS	May 2019
National Scholarship , PRC Ministry of Education	Nov 2018

Approved Proposals

- Wu, Q. (PI)**, Shen, Y. *A Local UV Benchmark for Interpreting Unobscured Coronal-Line AGNs*. 19 orbits with **HST** cycle 34. HST-GO-18341.
- Wu, Q. (PI)**, Shen, Y. *HST UV Spectroscopy of High-accretion-rate AGNs and the Origin of Offset in the Broad-Line Region Size-Luminosity Relation*. 19 orbits with **HST** cycle 31. HST-GO-17433.
- Wu, Q. (PI)**, Wu, J., Sai, H. *Long-Term Optical Monitoring on the Black Hole Binary MAXI J1820+070*. 120 hours with LCOGT from 2021 CTAP. CTAP2021-A0019.

Publications

Major contribution papers

- **Wu, Q.**, Shen, Y., Done, C., Goad, M. R., & Hagen, S. (2026). **Understanding the Broad-line Region of Active Galactic Nuclei with Photoionization. II. Slim disks, Self-shadowing, and BLR sizes**. arXiv preprint, arXiv:2606.00992.
- **Wu, Q.**, Shen, Y., Guo, H., et al. (2025). **Understanding the Broad-line Region of Active Galactic Nuclei with Photoionization. I The Moderate-Accretion Regime**. The Astrophysical Journal 980 (1), 134.
- **Wu, Q.**, & Shen, Y. (2023). **Improved Redshifts for DESI EDR Quasars**. Research Notes of the AAS, 7(9), 190.
- **Wu, Q.**, & Shen, Y. (2022). **A Catalog of Quasar Properties from Sloan Digital Sky Survey Data Release 16**. The Astrophysical Journal Supplement Series, 263(2), 42.
- Zheng, W. M., **Wu, Q.**, Wu, J., et al. (2022). **The Disk Veiling Effect of the Black Hole Low-mass X-Ray Binary A0620-00***. The Astrophysical Journal, 925(1), 83.
- **Wu, Q.**, Yu, H. R., Liao, S., Du, M. (2021). **Spin mode reconstruction in Lagrangian space**. Physical Review D, 103(6), 063522.

Other contributing-author papers

- Zhao, X., et al. (2026). **Transient Relativistic Iron Emission Line from an X-Ray Flaring Supermassive Black Hole**. The Astrophysical Journal, 1000(1), 28.
- Zhuang, M., et al. (2026). **NEXUS: A Spectroscopic Census of Broad-line AGNs and Little Red Dots at $3 \lesssim z \lesssim 6$** .

The Astrophysical Journal, 999(1), 31.

- Kollmeier, J. A., et al. (2026). Sloan Digital Sky Survey. V. Pioneering Panoptic Spectroscopy. The Astronomical Journal, 171(1), 52.
- Cheng, X., et al. (2025). Quest for a Coherent Definition of Weak-line Quasars and Its Physical Implications. The Astrophysical Journal, 994(2), 213.
- Nandra, K., et al. (2025). The eROSITA Final Equatorial Depth Survey (eFEDS)-The hard X-ray selected sample. Astronomy & Astrophysics, 693, A212.
- Waddell, S. GH, et al. (2024). The eROSITA Final Equatorial Depth Survey (eFEDS): Complex absorption and soft excesses in hard X-ray-selected active galactic nuclei. Astronomy & Astrophysics, 690, A132.
- Musiimenta, B., et al. (2023). A new discovery space opened by eROSITA-Ionised AGN outflows from X-ray selected samples. Astronomy & Astrophysics, 679, A84.
- Cheng, S., et al. (2020). CUBE-Towards an Optimal Scaling of Cosmological N-body Simulations. In 2020 20th IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGRID) (pp. 685-690). IEEE.
- Wen, J., et al. (2019). GRID: a student project to monitor the transient gamma-ray sky in the multi-messenger astronomy era. Experimental Astronomy (2019), 48, 77-95.

Talks

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- **WHU AGN/Galaxy Seminar**, "Linking AGN Accretion Flows and Broad-Line Region Properties through Photoionization Modeling", Wuhan University Jul 2026
 - **XMU BH seminar**, "Linking AGN Accretion Flows and Broad-Line Region Properties through Photoionization Modeling", Xiamen University Jul 2026
 - **UMich Galaxies Seminar**, "Understanding the AGN line-emitting regions: from optical spectral statistics to broad-line region photoionization modeling", UMich Feb 2026
 - **SDSS-V 2025 Collaboration Meeting**, "From Quasar Spectra to Structure: SDSS-V Quasar Spectral Analysis and AGN Photoionization Studies" (remote), MPIA Jun 2025
 - **UIUC Journal Club**, "Understanding the broad-line region of AGN with photoionization", UIUC Apr, 2025
 - **AGN surveys workshop**, "Understanding the broad-line region of AGN with photoionization", Munich Institute for Astro-, Particle and BioPhysics Mar 2025
 - **SDSS-V 2024 Collaboration Meeting**, "Quasar value-added catalog and visual inspection", New Mexico State University Jun, 2024
 - **Center for AstroPhysical Surveys seminars**, "Photoionization in quasar Broad Line Region", UIUC Aug, 2023
 - **SDSS-V 2023 Collaboration Meeting**, "AGN demographics", Simons foundation, NY Aug, 2023
 - **UIUC Journal Club**, "Quasar spectral decompositions from public surveys", UIUC May, 2023
 - **AGN demographics working group**, "Tutorials on PyQSOFit" (remote), MPE Nov, 2022
 - **The 23rd Guoshoujing Conference**, "Correlations between halo spins and primordial perturbations", Zhejiang University May, 2021
 - **Binary system seminar**, "Dynamical measurements on black hole binary system MAXI J1820+070", Nanjing University Nov, 2020

Outreach

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| Mentor at the UIUC Society for Equity in Astronomy mentoring program | 2022 Fall - 2024 Spring |
| Volunteer at the UIUC girls astronomy summer camp | 2022 Summer |
| President of Xiamen University Astronomical Club | 2018-2019 |
| Video editor and translator of Mufu Astronomy Team | 2018-2020 |